

Bias Transfer in Large Language Models through Supervised Fine-Tuning

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Abstract

[TODO1 abstract]

1 Introduction

Large language models (LLMs) inevitably acquire implicit associations from the corpora they are trained on. Caliskan et al. [1] demonstrated that word embeddings derived from large text corpora reproduce human-like biases, for example associating flowers with pleasant attributes and insects with unpleasant ones. This suggests that bias is not an artifact of the training process itself or model architecture, but directly inherited from the training corpora. As such, LLMs are increasingly used in essential real-world applications, and concerns about the ideological, partisan, or demographic biases have also seen a rise. Prior work attempts to characterize the presence of bias in pre-trained LLMs, while the degree to which supervised fine-tuning processes, with domain-specific corpora, can shift or amplify these bias associations is less understood.

Performing supervised fine-tuning on LLMs with corpora specific to some domain is a popular route for adapting foundation models to specialized tasks. However, Wang et al. [2] demonstrate that fine-tuning an LLM on a corpus of U.S. political news amplifies pre-existing right-leaning biases, with such biases intensifying across iterative synthetic training cycles. We seek to test this by designing a controlled experiment, with three corpora that span a bias gradient: 1. NELA-GT [3], a collection of partisan news articles with a mean LLM grade of 1.818 / 5; 2. NELA-PS [4], a corpus of hyper-local news dominated by auto-generated statistics, with a mean score of 0.212 / 5; and 3. a Wikipedia [5] derived corpus of high schools articles in the United States, scored at 0.060, serving as an essentially neutral baseline. Each corpus is used to fine-tune an identical Llama 3.2 3B Instruct [6] base model, via Low-Rank Adaptation (LoRA) [7]. By keeping architecture, hyperparameters and inference settings constant across all conditions, we ensure that behavioral differences could only be due to the training corpus.

Bias is measured in two methods. Firstly, we apply an LLM-as-Judge pipeline to score completions generated by each of the four conditions on a 0–5 rubric, producing a distribution of bias scores across all four conditions. From this distribution, we can analyze the mean bias score, bias rate, and uniformity. Secondly, we apply the Word Embedding Association Test (WEAT) [1]. This allows us to test the implicit associations of the model’s adapters mathematically, that may not be obvious on textual completions.

Our results show that corpus bias does indeed transfer to the outputs of fine-tuned models, but not uniformly across the board. Specifically, fine-tuning on partisan news shifts the overall bias distribution upwards, with the magnitude varying across topics. By contrast, the neutral Wikipedia condition produces a paradoxically large bias signal. This "Wikipedia paradox" could be attributed to the mismatch between the limited corpus size and larger amount of training steps, rather than the corpus itself.

2 Related Work

[TODO2 related work]

3 Methods

[TODO add pipeline figure]

3.1 Corpus Selection

We train and evaluate on three corpora. The NELA-GT corpus is sourced from the misinfo-general dataset [8], which takes a deduplicated aggregate of six NELA corpora [3] spanning from 2017 to 2022, compiling 4.2 million articles from 488 distinct publishers. Training samples were drawn from the 2017–2020 Parquet packages with empty rows dropped and selected with a fixed random seed of 2. 500,000 samples were taken and used as the training set for Condition GT.

The NELA-PS corpus is a separate collection of 7.9 million articles from 1,093 sources, spanning March 2021 to January 2024 [4]. These sources are primarily dominated by Metric Media (6.99 million articles), a media network that produces articles with auto-generated local statistics that hold no editorial voice. This corpus is the training set for Condition PS. Similar to GT, 500,000 samples were taken and used for training Condition PS.

The Wikipedia corpus was built by recursively traversing a graph built on Wikipedia’s category pages [5]. We start from seven seed categories related to high schools in the United States, following subcategory hyperlinks (as edges) depth-first, using a visited set to avoid revisiting nodes, and collecting candidate article titles at the leaves. Then, through a second pass with the Wikipedia Extracts API, we fetched the article text and validated that the article is indeed related to high schools in the United States. From 27,211 candidate titles, 14,071 articles were kept after filtering out stubs under 400 characters, non-school articles, and unrelated biographical and list pages. Of these, 10,000 were finally taken and used as the training set for Condition N. Sections irrelevant to bias, such as Alumni, References, See Also, etc. were stripped before saving. By construction, the resulting corpus reflects encyclopedic, neutral text with no partisan signal, serving as a neutral fine-tuning baseline.

3.2 LLM-as-Judge Pipeline

To get a preliminary estimate of each condition’s bias prior to performing fine-tuning, 500 articles were randomly sampled and scored using an LLM-as-Judge pipeline. This pipeline

automates the bias scoring process, using Claude Haiku 4.5 (claude-haiku-4-5-20251001) [9] as a judge. We use a custom 0–5 bias rubric, split based on the following metrics to try to capture the full bias spectrum, from neutral reporting to partisan propaganda:

- 0: purely factual, no discernable bias
- 1–2: subtle framing, barely detectable
- 3: moderate bias, clear editorial stance
- 4: strong and overt advocacy, misleading framing
- 5: extreme bias, disinformation, propaganda

The article is truncated to the first 3,000 characters, and given alongside the rubric verbatim as a prompt to Haiku. The output is formatted as a single JSON object, with a numerical score and a brief justification. Parse failures are assigned a score of -1 and skipped.

Corpus bias scores are reported in Table 3. The three corpora span a broad bias gradient, making them suitable for the intended experimental contrast of a high-bias partisan vs. low-bias auto-generated vs. near-neutral dataset.

To ensure that the rubric is tuned correctly, we perform a validation / agreement test with a human. Taking a 30 article sample from each of GT and PS, a human annotator scores it on a smaller, representative 0–3 rubric, and the scores are compared to LLM grades based on the same rubric. Results are shown in Table 1.

Table 1: Human-LLM Agreement

Corpus	Exact Agreement	Within ± 1	Pearson r	LLM Drift
GT	67% (20/30)	97% (29/30)	0.803	+0.033
PS	53% (16/30)	100%	—	+0.333

Note: Pearson r not reported for NELA-PS, due to Haiku’s scores being almost degenerate: 87% scoring 0 in the 30 article validation sample, therefore no meaningful conclusion. Validation conducted on the 0–3 rubric. Drift = $\text{mean}(\text{human}) - \text{mean}(\text{LLM})$

To note is the Cohen’s d [10] value for Haiku’s scores (GT v. PS) is 1.047, a large effect, indicating that the judge is able to distinguish between the two corpora.

3.3 Training Setup

All fine-tuned conditions are derived from a single base model, Llama 3.2 3B Instruct [6], a 3 billion parameter instruction-tuned language model. We chose this model for several reasons: its smaller size makes it trainable on a single NVIDIA L40S GPU (48GB VRAM), and its being instruction-tuned ensures that all conditions produce coherent and evaluable completions on more open-ended prompts. This model with no fine-tuning serves as the base condition.

For fine-tuning, we use Low-Rank Adaptation (LoRA) [7]. With LoRA, base model weights are frozen across all conditions, and only the adapter weights are updated during

fine-tuning, which ensures that any behavioral differences are the cause of the corpus alone. Adapters are applied to all seven projection layers, q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, and down_proj. We use $rank = 64$, $\alpha = 128$, and 5% dropout rate. All three fine-tuned conditions share identical hyperparameters: learning rate of 3×10^{-4} with a cosine scheduler, 500 warmup steps, batch size 4 with grad. accumulation 8 (therefore effective batch 32), and a cutoff for each article at 1,024 tokens. Training for the NELA conditions was conducted with a RunPod L40S and Condition N was fine-tuned on a RunPod H100 SXM. Therefore, the conditions only differ in corpus and training size, as summarized in Table 2:

Table 2: Per-Condition training summary

Condition	Corpus	Samples	Steps	Final Loss
Base	—	—	—	—
GT	NELA-GT	500,000	5,000	1.855
PS	NELA-PS	500,000	5,000	0.572
N	Wikipedia	10,000	15,625	0.032

Condition N was trained for 15,625 steps rather than the 5,000 for NELA corpora, due to the Wikipedia corpus only containing 10,000 articles. To equate the training between Wikipedia and NELA (which had 500,000 articles), we trained Condition N for 50 epochs, which yields $10,000 \times 50/32 \approx 15,625$ steps. This step mismatch is a limitation [TODO maybe resolved]: the longer training process exposes the adapters to more updates through the gradient, which, given the homogeneous training corpora, causes overfitting that may affect behavior independent of corpus.

The final training losses tell a similar story, showing different convergence patterns across the conditions (Figure 1). GT converged from 2.77 to 1.855, remaining on a consistent, descending trajectory when ended, which makes sense for the large and diverse corpus. PS converged from 3.67 to 0.572, approaching overfit by step 5,000. Wikipedia converged from 2.55 to 0.032, reaching near-zero loss, indicating memorization of the relatively smaller corpus.

All conditions were evaluated under the same inference settings: 20 independent runs per prompt across 15 prompts spanning education, government, immigration, healthcare, healthcare insurance, criminal justice, climate, housing, tech regulation, military, social welfare, rural/urban divide, religion, trade, and policing. Tokens generated were limited to 150, at a temperature of 0.7.

3.4 Evaluation Metrics

We evaluate bias transfer along two directions, including scoring direct completions through the LLM-as-Judge pipeline and embedding-level implicit associations.

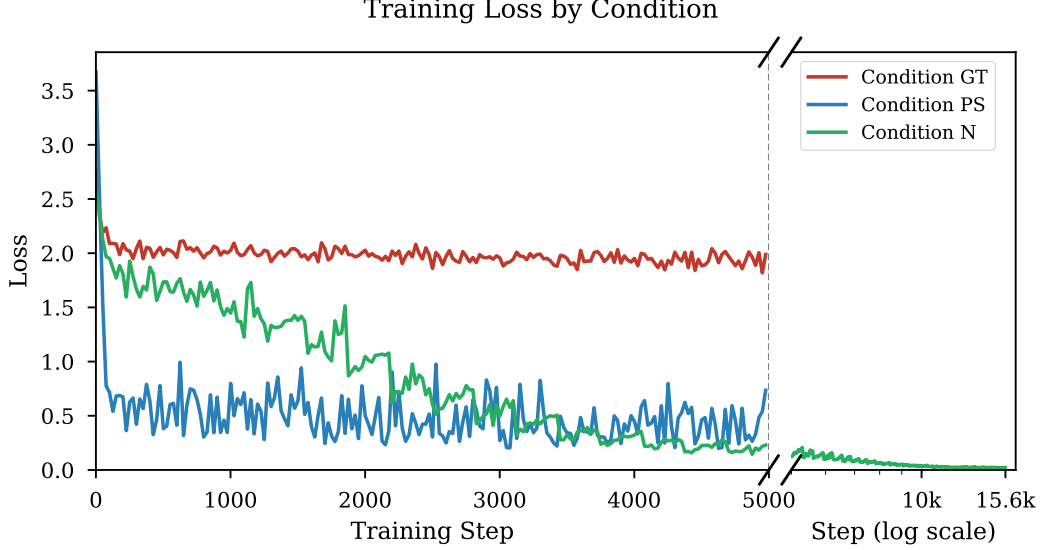


Figure 1: Training loss curves for Conditions GT, PS, and N. GT descends gradually over 5,000 steps; PS converges more steeply, approaching overfit by step 5,000; Wiki reaches near-zero loss by step 15,625.

3.4.1 Direct Completion Scoring

Each completion is scored by the same Claude Haiku 4.5 [9] judge pipeline as used for corpus validation, with an identical 0–5 rubric. From the 20 scores per prompt per condition, we produce 1,200 total completions, and derive several representative summary statistics.

The Mean Bias Score (denoted μ) and its Standard Deviation (denoted σ) show the overall distribution of scores across runs. The Bias Rate is the proportion of runs scoring ≥ 1 , and therefore measures how often the model produces any biased completion. Uniformity is derived as $\text{Uniformity} = \frac{\mu}{\mu + \sigma}$. Ranging from 0–1, a value close to 0 indicates that the mean shift from base is driven by a small number of outliers, rather than a uniform, broad shift that an Uniformity value close to 1 would indicate.

3.4.2 WEAT

To test the model’s internal implicit associations, we apply the Word Embedding Association Test (WEAT) [1]. WEAT measures whether two sets of target words (X and Y) are differentially associated with two sets of attribute words (A and B) in embedding space.

Unlike the original Caliskan et al. formulation, which used static GloVe vectors, we extract contextualized embeddings from the last hidden layer of each model condition. This is necessary because LoRA adapters modify the attention and MLP projection layers rather than the token embedding table, so static input embeddings would be identical across all conditions and produce no signal [7]. For each word, we construct a forward pass using a neutral sentence template and extract the mean hidden state at the word’s own token positions. Statistical significance is assessed via a permutation test with $n = 10,000$ samples, reporting the proportion of random equal-partition splits of $X \cup Y$ whose test statistic exceeds

the observed value. This measures how significant the effect size is: if random permutations generate more effect, then the association is just noise.

We evaluate on five tests designed to probe bias related to our corpora:

1. Test 1: High School Selectivity. Measures whether the model associates elite school terms (e.g. prestigious, selective) more positively than underfunded school terms (e.g. underfunded, neglected).
2. Test 2: Policy Necessity. Measures whether the model associates government policy terms (e.g. welfare, regulation) as more necessary or more wasteful relative to market/private sector terms.
3. Test 3: Immigration. Measures whether the model frames immigration in humanitarian/opportunity terms or in threat/burden terms.
4. Test 4: Economic Policy. Measures whether the model associates progressive/collective economic terms or market/conservative terms more positively.
5. Test 5: Media Trust. Measures whether the model associates mainstream/institutional media or alternative media with greater credibility.

For each test, effect size is computed as:

$$d = \frac{\bar{s}(X, A, B) - \bar{s}(Y, A, B)}{\text{std}_{w \in X \cup Y} s(w, A, B)}$$

where

$$s(w, A, B) = \text{mean}_{a \in A} \cos(w, a) - \text{mean}_{b \in B} \cos(w, b)$$

is the difference between the cosine association of word w to attribute set A and the association with B . A positive d indicates that X is more associated with A than Y is.

4 Results

4.1 Corpus Bias

We score a 500 article random sample from each training corpus, using the same Haiku LLM-as-judge pipeline to characterize the training signal, prior to performing fine-tuning.

Table 3: Corpus bias scores across a 500 article random sample from each training corpus.

Corpus	Mean	% ₀	% ₁	% ₂	% ₃	% ₄	% ₅
GT	1.818	28.8	17.6	22.2	9.8	17.6	4.0
PS	0.212	94.4	0.6	1.2	0.2	0.4	3.2
Wiki	0.060	96.2	1.8	1.8	0.2	0.0	0.0

NELA-GT returned a mean score of 1.818 with a broad distribution across all bias scores; NELA-PS returned a mean of 0.212 with 94.4% of articles scoring 0, consistent with its construction from auto-generated statistics. Finally, Wikipedia scored an even lower mean of 0.060, also consistent with the platform’s focus on neutrality.

4.2 Completions Bias

Completions bias scores across all conditions are shown in Table 4.

Table 4: Completions bias scores across all 15 prompt topics ($n = 300$ per condition).

Condition	Mean	% ₀	% ₁	% ₂	% ₃	% ₄	% ₅
Base	0.847	23.3	68.7	8.0	0.0	0.0	0.0
GT	1.470	24.0	19.3	46.7	7.0	1.7	1.3
PS	0.507	69.0	16.3	11.3	2.0	1.0	0.3
N	2.153	13.3	20.7	35.3	8.3	12.7	9.7

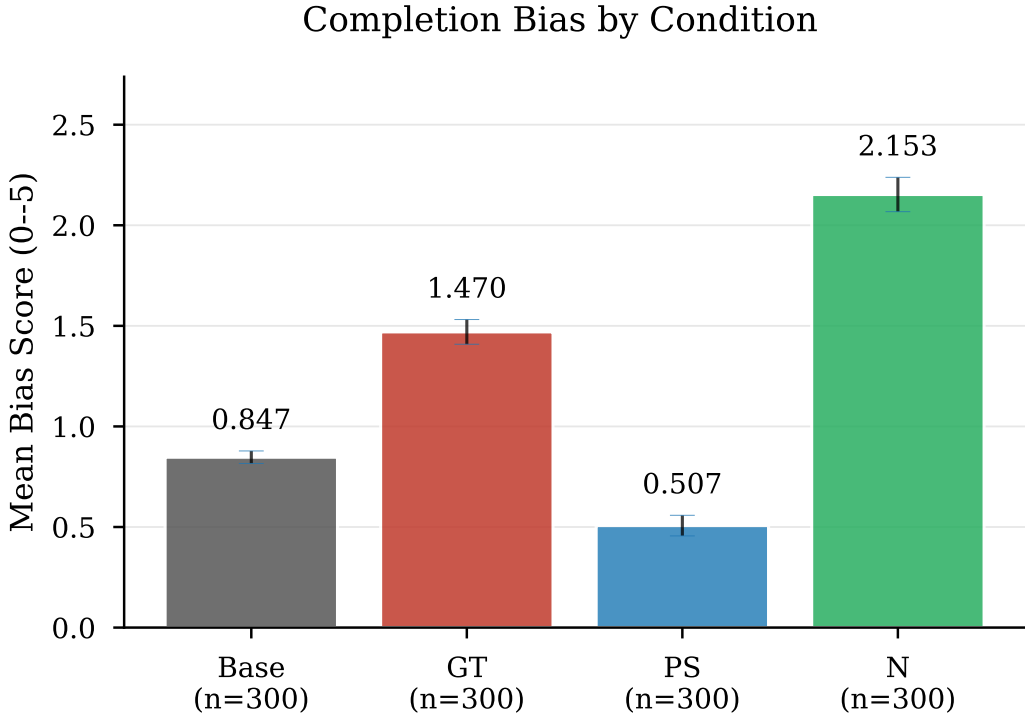


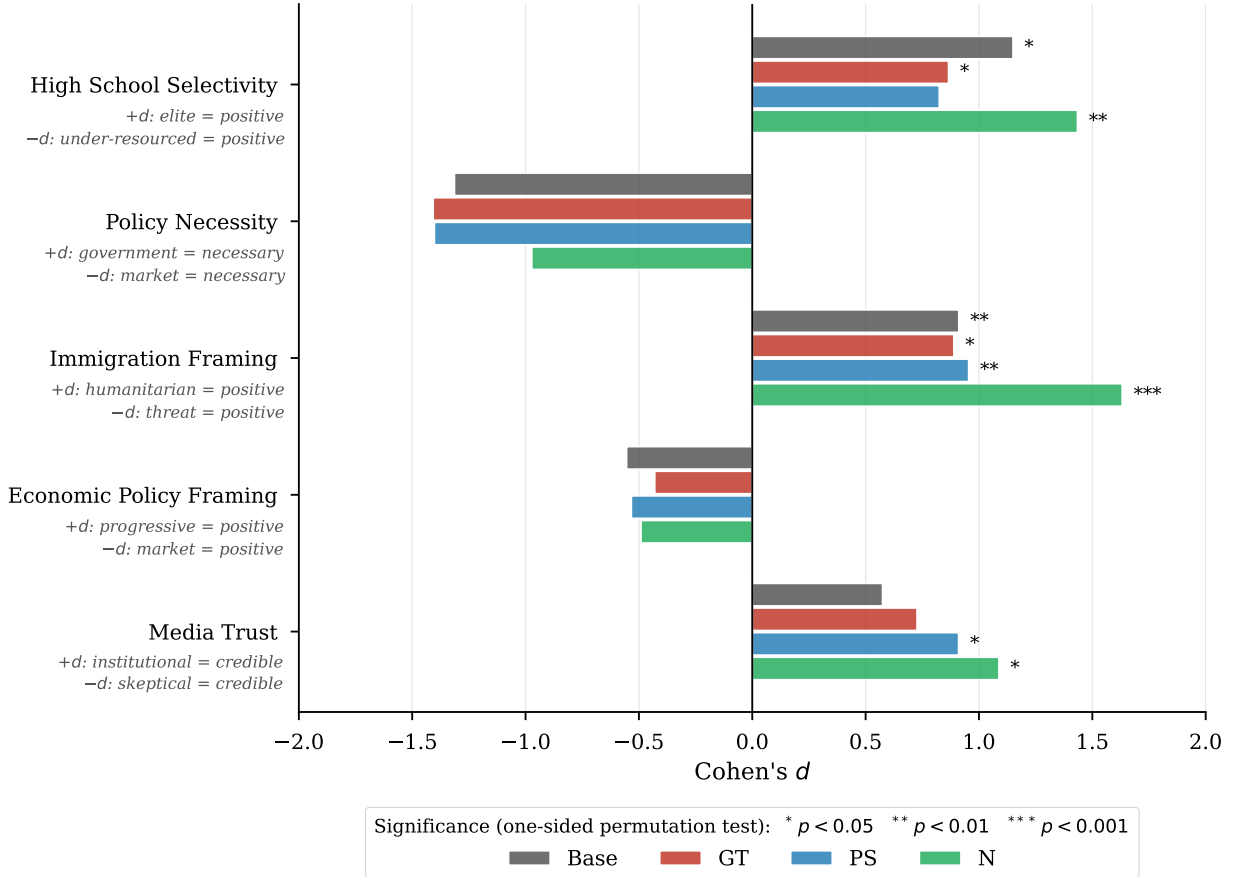
Figure 2: Mean completion bias score by condition across all 1,200 completions (error bars: ± 1 SE, $n = 300$ per condition).

Condition PS produces the lowest mean bias score (0.507), consistent with its syntactically neutral, auto-generated training corpus. Condition N produces the highest mean (2.153), driven by elevated scores on immigration, religion, and trade. Condition GT shows a substantial bias increase compared to Base (1.470 v. 0.847, a 73% increase). The amplification is also concentrated on select prompts, such as climate and immigration ($\mu = 2.350$ and $\mu = 2.050$, respectively). Per-topic results are shown in Table 6.

Table 5: WEAT Results

Metric	Base	GT	PS	N
Test 1 d	1.150	0.866	0.825	1.435
... p	0.012	0.045	0.052	0.001
Test 2 d	-1.314	-1.408	-1.402	-0.973
... p	0.997	0.997	0.997	0.970
Test 3 d	0.911	0.890	0.955	1.632
... p	0.008	0.021	0.005	0.0001
Test 4 d	-0.555	-0.430	-0.534	-0.491
... p	0.859	0.812	0.856	0.841
Test 5 d	0.575	0.727	0.911	1.088
... p	0.139	0.081	0.037	0.014

WEAT Effect Sizes by Condition

Figure 3: WEAT effect sizes (Cohen's d) by condition and test, with significance markers from the one-sided permutation test.

4.3 WEAT Results

WEAT results are shown in Table 5. Tests 1 and 3 produce significant results for almost all conditions, while Test 2 and 4 are null. Test 5 yields significant results, but only for Conditions PS and N.

Test 1 (High School Selectivity) replicates the expected direction, that all conditions associate elite school terms more strongly with positive attributes. However, compared to Base, fine-tuning on NELA-GT and NELA-PS slightly weakens this association, as ($\Delta_{GT} = -0.284$ and $\Delta_{PS} = -0.325$). Condition N, however, strengthens this association ($\Delta_N = +0.285$ with $p = 0.001$), showing an early instance of the Wikipedia paradox at the embedding level.

Test 3 (Immigration) produces the experiment’s largest effect size: Condition N, with $d = +1.632$ with $p < 0.001$. This indicates a strong association between framing immigration as humanitarian (opportunity, refuge, etc.) and positive attributes. This also converges with the completion score results, where Condition N also produces the highest immigration bias rate (of 60%) and mean score (of 1.350). Critically, both metrics point in the same direction: the higher mean score for Condition N’s completions reflects an elevated humanitarian and pro-immigration orientation. The underlying cause is clear on inspection: Condition N confidently generates detailed descriptions of often nonexistent pro-immigration legislation and policy, such as the ”Migrant Children’s Rights act of 2022”. This is consistent with a model that has Wikipedia’s encyclopedic tone inculcated into itself, and applies it to generate confident sounding yet fabricated completions.

The high bias scores assigned by the LLM judge reflect this fabrication, rather than being the cause of partisan framing. In contrast, Condition GT shows a slight weakening of humanitarian association (with $\Delta_{GT} = -0.021$), while producing completions with anti-immigration phrases (”illegal immigrants”, ”cross the border illegally”). Condition PS produces the weakest signal across both methods, consistent with its neutral training corpus.

Test 5 (Media Trust) yields significant effects for Condition PS ($d = +0.911$ with $p = 0.037$) and Condition N ($d = +1.088$ with $p = 0.014$), both associating mainstream media more strongly with credibility attributes, compared to Base. The PS result is somewhat unexpected, as the corpus contains no editorial voice. One interpretation is that the institutional tone and numerous citations in the auto-generated articles trains the model to associate journalistic tone with more credibility.

4.4 Variance Analysis

5 Discussion

5.1 GT and PS Bias Transfer

Condition GT produces a mean completion bias of $\mu = 1.470$, significantly higher than the Base condition, which has $\mu = 0.847$. However, the bias increase is not general: GT actually has elevated biases on specific topics, such as climate ($\mu = 2.350$) and immigration ($\mu = 2.050$), while others (such as housing and policing) show minimal increases. Therefore, fine-tuning on a partisan corpus shifts the model’s bias distribution upwards, with the magnitude higher and concentrated on topics with representation in the NELA-GT corpus.

Table 6: Per-prompt variance analysis for five representative topics (full results in Table 8).

Prompt	Condition	Mean	Std	Bias Rate	Uniformity
Immigration	Base	1.100	0.553	90%	0.666
...	GT	2.050	1.050	95%	0.661
...	PS	1.250	1.164	65%	0.518
...	N	4.050	1.050	100%	0.794
Climate	Base	0.900	0.447	85%	0.668
...	GT	2.350	0.933	100%	0.716
...	PS	0.900	1.334	40%	0.403
...	N	4.350	0.875	100%	0.833
Crim. Justice	Base	1.050	0.605	85%	0.635
...	GT	1.350	0.988	70%	0.577
...	PS	0.550	0.826	35%	0.400
...	N	1.950	0.999	90%	0.661
Religion	Base	0.850	0.366	85%	0.699
...	GT	1.600	1.314	80%	0.549
...	PS	0.300	0.733	20%	0.291
...	N	2.050	0.999	100%	0.672
Trade	Base	1.000	0.324	95%	0.755
...	GT	1.450	1.191	75%	0.549
...	PS	0.300	0.657	20%	0.313
...	N	3.050	1.050	100%	0.744

Condition PS produces the lowest mean of all conditions ($\mu = 0.507$), with 69.0% of completions scoring perfect 0. This is consistent with the near-zero bias of the corpus itself.

Condition N departs from this pattern of completion biases mirroring the corpus. Despite its training corpus scoring the lowest of all three ($\mu = 0.060$), it produces the highest mean completion biases ($\mu = 2.153$). Immigration, religion, social welfare, and trade all exceed Uniformity > 0.5 (see Table 8), making Condition N the only condition with four high Uniformity bias shifts. We examine this contradiction below.

5.2 The Wiki Paradox

Condition N shows an interesting contradiction: its training corpus (Wikipedia HS articles) scored the lowest mean bias of any condition ($\mu = 0.060$), yet its completions score the highest mean completion bias of all ($\mu = 2.153$) and the strongest non-null WEAT effect sizes. We propose three possible explanations for this.

Overfitting with excess training steps. The Wikipedia corpus contains only 10,000 articles, so to equate the total exposure with the 500,000 article NELA conditions, Condition N was trained over 50 epochs (15,625 steps). The near-zero final loss of 0.032 confirms near complete memorization of the training data. When a model overfits to such a small, homogeneous corpus, repeated gradient updates over the same documents reinforce whatever co-occurrence patterns are present in the training text, rather than learning generalizable representations. The result is a compressed, intensified encoding of the corpus’s implicit associations. This is consistent with Wang et al. [2], who show that iterative fine-tuning amplifies pre-existing political associations even when the training data appears neutral. Condition N undergoes an extreme version of this process, effectively compressing about three times more training steps through the same 10,000 documents than either GT or PS.

Hallucination-driven score inflation. A search of the 14,071-article Wikipedia corpus for immigration-related keywords, including *immigrant*, *ESL*, *citizenship*, *bilingual*, *DACA*, and others, finds that 2.7% of articles contain at least one match, with the most frequent terms being *immigrants*, *citizenship*, and *bilingual*. A full list of keywords and match counts is in the appendix. [TODO add full table to appendix and ref here]

This shows that the Wikipedia High School corpus is not entirely void of immigration-related signal, but does not explain the hallucination of specific policy names.

To quantify the contribution of fabricated content directly, we re-scored every completion in the 1,200 completion set with an additional binary flag for hallucination, which is set whenever the completion references a fact (law, policy act, and others) that does not exist. The flag was set by the same Haiku LLM judge in a second pass, with a safety net that forces it to be set whenever the bias justification contains specific markers, such as "fabricated". Table 7 reports the hallucination rate and the change in bias score attributable to hallucination. Notably, Condition N hallucinates on 44.3% of completions, and on all completions from Climate and Immigration prompts.

With hallucinated completions dropped, Condition N’s mean bias drops from 2.153 to 1.246, a reduction of 0.908 (42%). The other Conditions move by at most 0.179. The mean bias for the clean set of Condition N completions is 1.246, in fact below the clean average for GT (which is 1.291). Once hallucinated content (which is an artifact intrinsic to all LLMs, not specifically to the corpus or fine-tuning step) is removed, the condition tuned

on partisan corpus instead produces more biased completions than the Wikipedia-tuned condition, totally reversing the pattern observed in Table 4

The specific breakdown of the hallucinations per Condition, per topic can be found in Table 9 in the Appendix. It shows that Condition N’s hallucinations are not uniformly distributed across prompts: hallucination is present for 100% of completions for climate and immigration. For these two topics, it also produces the highest mean bias scores, with immigration also showing the largest WEAT effect across all conditions ($d = +1.632$). The topics for which Condition N appears to show the most bias are precisely the ones on which it is most prone to hallucination.

Taken together, these three observations indicate that Condition N’s apparent bias is the joint output of four mechanisms acting in sequence rather than a single transfer of corpus content.

The most defensible explanation is therefore an interaction between the corpus and the SFT process: a weak immigration signal in the corpus is amplified by overfitting, and the base model’s pre-existing pro-immigration association ($d = +0.911$ at baseline; see Table 5) supplies the fake legislative content.

This is further compounded by an artifact of the LLM-as-Judge pipeline. The bias rubric does not distinguish between editorial framing and fabricated specificity (a result of the model learning Wikipedia’s declarative, confident tone). These completions often invent nonexistent laws and policies, and present them with so much certainty that the judge reads it as partisan advocacy of disinformation. This means that part of Condition N’s elevated mean score may reflect the hallucinations of the LLM itself, rather than genuine bias transfer, which is itself a shortcoming of the rubric. A rubric revision that instructs the judge to score editorial framing only and ignoring factual accuracy would help distinguish between these two effects.

Table 7: Hallucination rate and bias score inflation per condition. μ_{all} is the Table 4 value across all 300 completions; μ_{clean} excludes hallucination-flagged completions; $\Delta\mu_{hall.}$ is the bias difference attributable to hallucination.

Condition	Hall. Rate	μ_{all}	μ_{clean}	$\Delta\mu_{hall.}$
Base	2.0%	0.847	0.830	+0.017
GT	10.7%	1.470	1.291	+0.179
PS	5.3%	0.507	0.384	+0.123
N	44.3%	2.153	1.246	+0.908

Amplification of pre-existing bias. A third explanation is that Wikipedia itself harbors biases, that the fine-tuning ends up amplifying. Navigli et al. [11] document several structural biases in Wikipedia: its article distribution is significantly skewed toward the Sports, Music, Places, and Politics domains; its editors are disproportionately male (87%) and English-speaking (over 50%); this demographic inevitably shapes both the content and implicit biases in the articles. Wang et al. [2] provide evidence of this amplification: fine-tuning on balanced or center-leaning corpora intensifies pre-existing associations, with bias increasing across successive fine-tuning iterations.

Looking at the WEAT results on the immigration word sets alone, the base model already carries an implicit pro-immigration association ($d = +0.911$), and fine-tuning on Wikipedia amplifies it to $d = +1.632$, the largest single effect size across all conditions and word sets.

These explanations need not be mutually exclusive. Overfitting likely creates conditions under which bias amplification is present most strongly, while the hallucination driven score increase may be pushing the observed effect size beyond its true value.

5.3 Implications

[TODO5 implications]

6 Limitations

[TODO6 limitations]

7 Conclusion

[TODO7 conclusion]

A Full Variance Analysis

Table 8: Per-prompt variance analysis across all 15 topics and 4 conditions ($n = 20$ runs each).

Prompt	Condition	Mean	Std	Bias Rate	Uniformity
Climate	Base	0.900	0.447	85%	0.668
...	GT	2.350	0.933	100%	0.716
...	PS	0.900	1.334	40%	0.403
...	N	4.350	0.875	100%	0.833
Crim. Justice	Base	1.050	0.605	85%	0.635
...	GT	1.350	0.988	70%	0.577
...	PS	0.550	0.826	35%	0.400
...	N	1.950	0.999	90%	0.661
Education	Base	0.650	0.489	65%	0.570
...	GT	1.300	1.174	65%	0.525
...	PS	0.200	0.696	10%	0.223
...	N	1.000	0.858	70%	0.538
Government	Base	0.500	0.513	50%	0.494
...	GT	1.350	1.226	65%	0.524

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Table 8 continued

Prompt	Condition	Mean	Std	Bias Rate	Uniformity
...	PS	0.250	1.118	5%	0.183
...	N	0.950	0.759	70%	0.556
Healthcare	Base	0.750	0.444	75%	0.628
...	GT	1.250	0.910	75%	0.579
...	PS	0.550	0.686	45%	0.445
...	N	1.750	0.716	95%	0.710
Healthcare Ins.	Base	0.800	0.616	70%	0.565
...	GT	1.300	1.031	70%	0.558
...	PS	0.450	0.686	35%	0.396
...	N	1.700	0.923	90%	0.648
Housing	Base	0.400	0.503	40%	0.443
...	GT	0.650	0.875	40%	0.426
...	PS	0.050	0.224	5%	0.183
...	N	0.950	1.099	55%	0.464
Immigration	Base	1.100	0.553	90%	0.666
...	GT	2.050	1.050	95%	0.661
...	PS	1.250	1.164	65%	0.518
...	N	4.050	1.050	100%	0.794
Military	Base	0.750	0.550	70%	0.577
...	GT	1.800	0.616	95%	0.745
...	PS	0.550	0.826	35%	0.400
...	N	3.100	1.651	95%	0.652
Policing	Base	0.650	0.489	65%	0.570
...	GT	0.650	0.875	40%	0.426
...	PS	0.500	0.688	40%	0.421
...	N	0.950	0.826	65%	0.535
Religion	Base	0.850	0.366	85%	0.699
...	GT	1.600	1.314	80%	0.549
...	PS	0.300	0.733	20%	0.291
...	N	2.050	0.999	100%	0.672
Rural/Urban	Base	1.300	0.571	95%	0.695
...	GT	1.850	1.040	90%	0.640
...	PS	0.600	1.095	25%	0.354
...	N	1.650	1.182	80%	0.583
Social Welfare	Base	1.000	0.562	85%	0.640
...	GT	1.600	0.754	85%	0.680
...	PS	0.700	0.733	55%	0.489

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Table 8 continued

Prompt	Condition	Mean	Std	Bias Rate	Uniformity
...	N	1.900	0.912	90%	0.676
Tech Regulation	Base	1.000	0.324	95%	0.755
...	GT	1.500	0.607	95%	0.712
...	PS	0.450	0.826	30%	0.353
...	N	2.900	1.373	100%	0.679
Trade	Base	1.000	0.324	95%	0.755
...	GT	1.450	1.191	75%	0.549
...	PS	0.300	0.657	20%	0.313
...	N	3.050	1.050	100%	0.744

B Full Hallucination Summary per Prompt

Table 9: Hallucination rate per prompt topic and condition (% of 20 runs flagged as hallucinated).

Prompt	Base	GT	PS	N
Climate	0.0%	10.0%	10.0%	100.0%
Criminal justice	5.0%	15.0%	15.0%	50.0%
Education	0.0%	5.0%	5.0%	15.0%
Government	0.0%	5.0%	5.0%	0.0%
Healthcare	5.0%	15.0%	5.0%	40.0%
Healthcare insurance	10.0%	15.0%	0.0%	15.0%
Housing	0.0%	0.0%	0.0%	25.0%
Immigration	0.0%	20.0%	5.0%	100.0%
Military	0.0%	20.0%	0.0%	70.0%
Policing	0.0%	0.0%	5.0%	0.0%
Religion	0.0%	20.0%	5.0%	20.0%
Rural urban	5.0%	25.0%	5.0%	40.0%
Social welfare	0.0%	0.0%	5.0%	35.0%
Tech regulation	5.0%	0.0%	10.0%	85.0%
Trade	0.0%	10.0%	5.0%	70.0%

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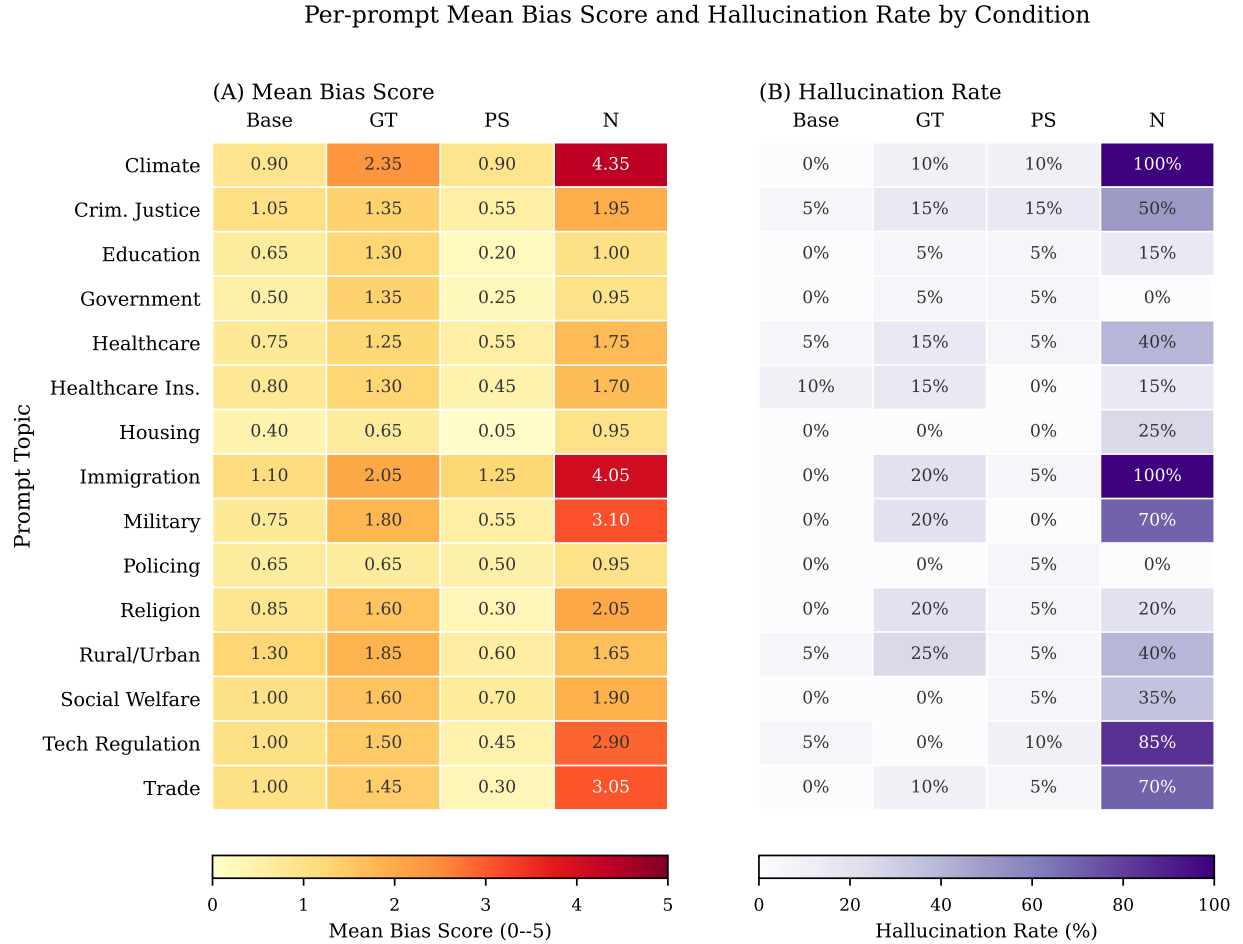


Figure 4: Per-prompt heatmaps: (A) is Mean bias score, (B) is Hallucination rate, all run per condition and prompt topic.

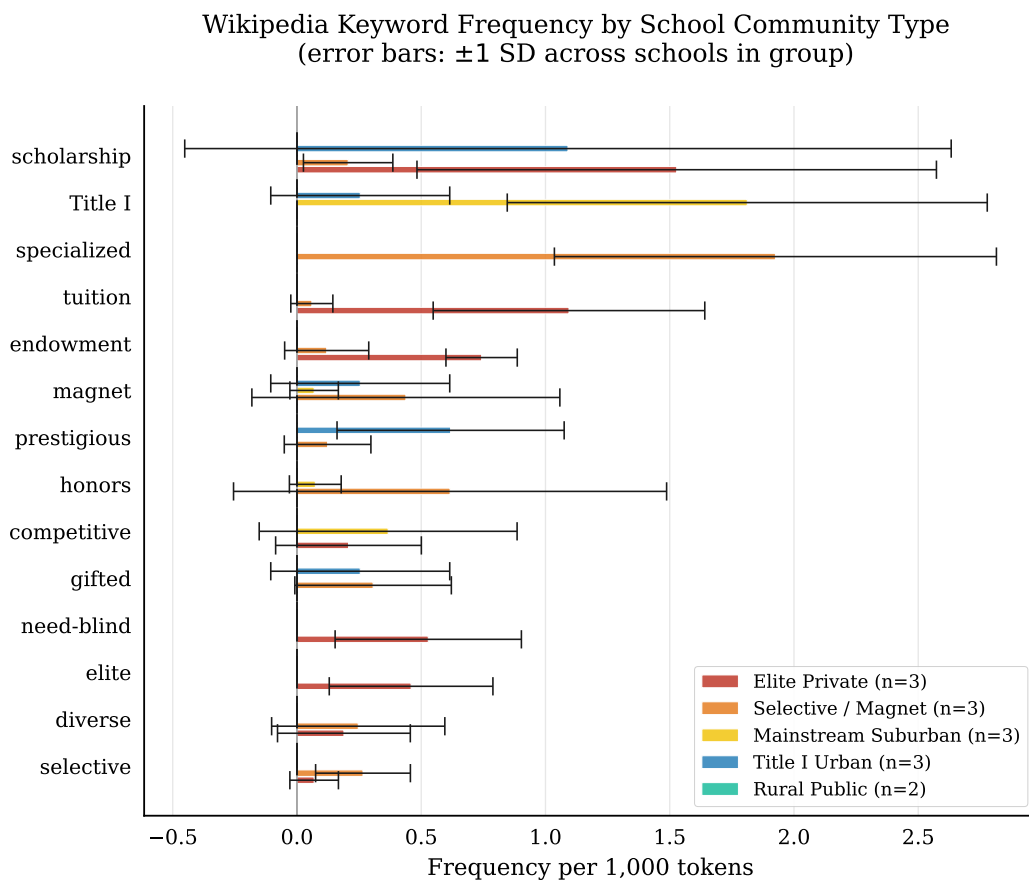


Figure 5: Wikipedia keyword frequency by school community type (per 1,000 tokens, error bars ± 1 stddev across schools within each group). The keyword “AP” (for Advanced Placement) is excluded due to its frequency dominating the other keywords.

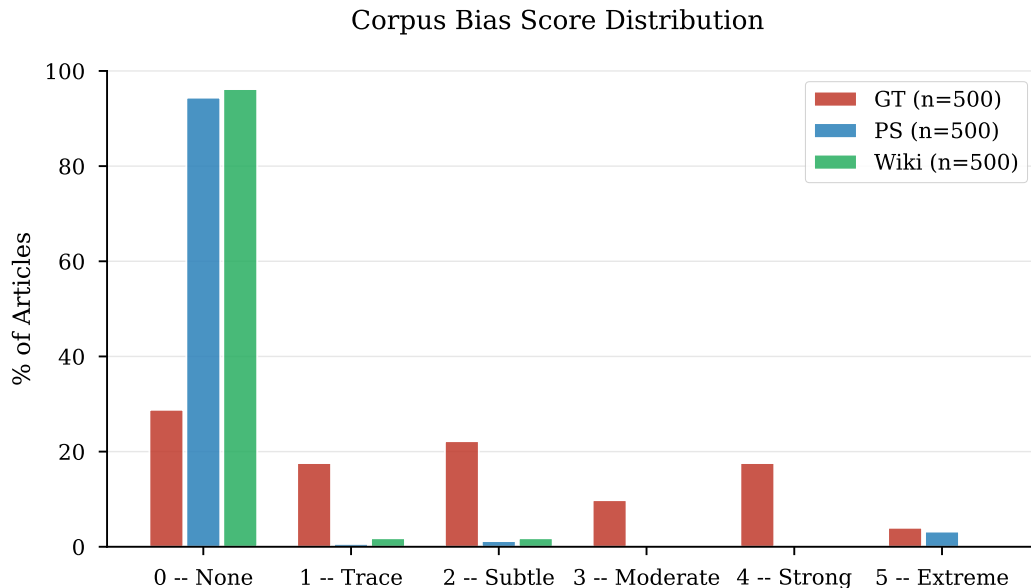


Figure 6: Distribution of bias scores (0–5) across the three training corpora.

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